

kinds of structures, to mine and process it,” explains Singhal.

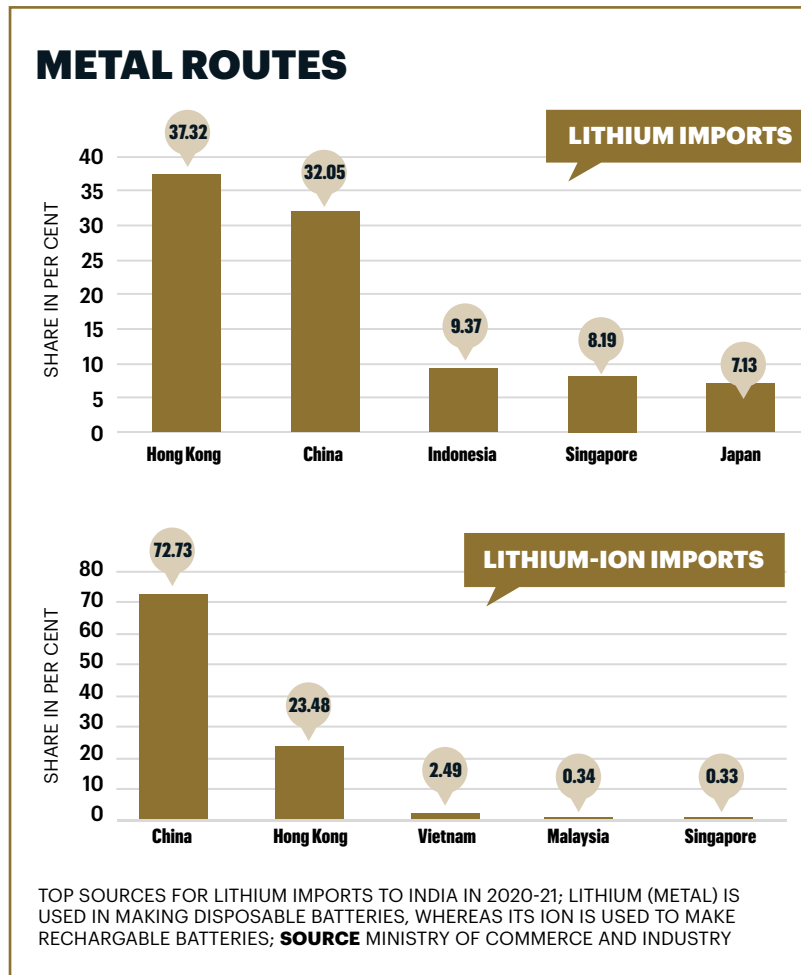
Another question that remains to be answered is regarding who will finally bid and mine the reserves. The reason: lithium and seven other metals, including beryllium and zirconium, fall under the restricted list of the Department of Atomic Energy that forbids private players from mining or producing them. Not only that, the average total cash cost of operating 11 hard-rock lithium producers was pegged at \$2,540 per tonne of LCE in 2019, by S&P Global. Some experts estimate this cost to be upwards of \$2,500-3,000 these days, excluding capital investments.

And as India bets big on the recent discovery, there is yet another opportunity it can leverage, which is recycling end-of-life lithium batteries. Industry experts believe that in the next two years, 90 per cent of the lithium mined today will end up in the lithium-ion batteries used in EVs. This is where home-grown recycling companies like Attero Recycling, Gravita India and Metastable Materials come in.

CATCHING UP

The uptake in sales of EVs and green energy-storage solutions has pushed the demand for lithium to astronomical heights. And as China controls 90 per cent of this market, there is sustained geopolitical push from countries like the US, the EU and even India to break China’s stronghold. “The US has come up with a new law called the Inflation Reduction Act, and it’s a \$500-billion subsidy. This states that EV subsidies will be given to those automobile OEMs who can prove that the batteries or the EV is not coming from China. It’s either domestic or coming from friendly countries,” says Gupta.

And as lithium emerges as a



key element for the future, India isn’t settling down with the J&K reserves. For instance, the GSI has carried out 14 exploration projects for lithium and associated elements across states between 2016-17 and 2020-21, and another five such projects in 2021-22. Moreover, the Atomic Minerals Directorate (AMD) has been tasked to carry out exploration projects for lithium in parts of Karnataka and preliminary surveys have indicated the presence of lithium resources of 1,600 tonnes (inferred) in Mandya, Karnataka. And as India lacked the two primary raw materials—lithium and cobalt—a joint venture

between National Aluminium Company Ltd, Hindustan Copper Ltd and Mineral Exploration Corporation Ltd named Khanij Bidesh India Ltd (Kabil) was formed in 2019 to supply strategic minerals from overseas locations to industries in India. Hopefully, India will discover more lithium reserves, while speeding up the process of mining the reserves already discovered. This will not only secure lithium supplies for the domestic market, but also help in building international bridges through exports. **BT**

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